Times = [7,13,22,45,100,250], K = 20, *best_T_sum_brute*(Times, K, T1), *best_T_sum_hist*(Times, K, T2), T1 = T2.

K = 20,
T1 = T2, **T2** = 13,
Times = [7, 13, 22, 45, 100, 250]

Next | 10 | 100 | 1,000 | Stop

Times = [2,5,8,12,15,18,22,25,28,32,35,38], K = 10, *build_histogram*(Times, K, Hist).

Hist = [0, 0, 4, 0, 0, 4, 0, 0, 4, 0],
K = 10,
Times = [2, 5, 8, 12, 15, 18, 22, 25, 28, 32, 35, 38]

Next | 10 | 100 | 1,000 | Stop

Times = [2,5,8,12,15,18,22,25,28,32,35,38], K = 10, *sum_wait_times*(Times, K, 5, Sum), *max_wait_times*(Times, K, 5, Max).

K = 10,
Max = 7,
Sum = 40,
Times = [2, 5, 8, 12, 15, 18, 22, 25, 28, 32, 35, 38]

optimal_first_bus([123], 500, T_sum, T_max).

T_max = T_sum, **T_sum** = 123

0.102 seconds cpu time

Next | 10 | 100 | 1,000 | Stop

optimal_first_bus([10,20,30], 100, T_sum, T_max).

T_max = T_sum, **T_sum** = 30

Next | 10 | 100 | 1,000 | Stop

optimal_first_bus([5,10,15,20], 5, T_sum, T_max).

T_max = T_sum, **T_sum** = 0

Next | 10 | 100 | 1,000 | Stop

Times = [2,5,8,12,15,20,25,30,35,40], K = 10, *optimal_first_bus*(Times, K, T_sum, T_max).

K = 10,
T_max = 2,
T_sum = 5,
Times = [2, 5, 8, 12, 15, 20, 25, 30, 35, 40]

Next | 10 | 100 | 1,000 | Stop

optimal_first_bus([7,7,7,7], 10, T_sum, T_max).

T_max = T_sum, **T_sum** = 7

Next | 10 | 100 | 1,000 | Stop

optimal_first_bus([1,2,3,4,5,6,7,8,9,10], 10, T_sum, T_max).

T_max = T_sum, **T_sum** = 0

Next | 10 | 100 | 1,000 | Stop